North Hollywood High School

Dynamic Planet C Test

CSE 2025

/90

Instructions:

- Do not open the test until time starts.
- You are permitted a binder and a calculator (without internet access).
- You are required to show all your work!
- All numerical answers should be reported to appropriate significant digits.
- Good luck!

Section 1: Plate Tectonics (10)

1. Which of the following features cannot form at a divergent plate boundary? (1)
    a. Mid-ocean ridge
    b. Volcanoes
    c. Rift valley
    d. Trench

2. Which of the following correctly matches the landform to its stage in the Wilson Cycle? (1)
    a. Red Sea, Embryonic
    b. Pacific Ocean, Declining
    c. Mediterranean, Suturing
    d. Atlantic, Declining

3. Which of the following is true about plate motion? (1)
    a. Slab suction is the driving force of motion
    b. Basal drag only resists plate motion
    c. Plates on average move around 1-2 meters per year
    d. Plate motion along mid ocean ridges are around equal on both sides

4. Explain what a fracture zone is and the difference between it and a transform fault. (3)

5. How do marine magnetic anomalies on the seafloor prove the existence of seafloor spreading? (2)

Section 2: Marine Provinces (10)

1. Which of the following features indicate a passive margin? (1)
    a. Continental shelf
    b. Continental slope
    c. Continental rise
    d. Trench

2. What rock commonly makes up turbidite deposits? (1)
    a. Greywacke
    b. Breccia
    c. Limestone
    d. Sandstone

3. Which of the following cover the majority of Earth's surface? (1)
    a. Continental rises
    b. Abyssal plains
    c. Mid ocean ridges
    d. Continents

4. What types of sediments are most common from hydrothermal vents?
    a. Terrigenous
    b. Biogenous
    c. Hydrogenous
    d. Cosmogenous

5. Explain the difference and formation of seamounts and guyots. (4)

6. Below is an image of the Hawaiian Islands. With the knowledge that the largest island is the youngest, explain the formation of these islands and how you can deduce the direction of movement of the tectonic plate. (3)



Section 3: Seawater (9)

1. Which of the following is closest to the salinity of seawater? (1)
    a. 0.035%
    b. 0.7%
    c. 3.5%
    d. 7%

2. Which of the following have the lowest concentration in seawater? (1)
    a. Potassium
    b. Sulfate
    c. Magnesium
    d. Nitrogen

3. Which of the following latitudes have the highest salinity? (1)
    a. 0
    b. 15 N
    c. 30 N
    d. 45 N

4. If I mix two parcels of water together with the same density but different temperatures and salinities, how will the density of the mixed parcel compare to the original parcels? Why is this? (3)

5. Deep ocean water contains more CO_2 than surface water, but it is not highly acidic. Why does deep-ocean water contain more CO_2 and why is it not highly acidic? (3)

Section 4: Waves (10)

1. Which of the following does not contribute to wave speed? (1)
    a. Fetch distance
    b. Time of day
    c. Wind speed
    d. Water depth

2. Beaufort Wind Scale describes sea surface conditions by which of the following? (1)
    a. Wind speed
    b. Wave frequency
    c. Wave height
    d. Wave speed

3. Which of the following are necessary for the formation of a rogue wave? (1)
    a. Stormy weather
    b. Earthquakes
    c. High tides
    d. Wave interference

4. There is a tsunami with a depth of 90 meters. What is the speed of this tsunami in meters per second? Is this a realistic speed for a tsunami in the ocean? What about near shore? (4)

5. How are internal waves formed and where are they naturally found? (3)

Section 5: Ocean Circulation (13)

1. About what percentage of wind energy is transferred to the ocean through friction? (1)
    a. 1%
    b. 2%
    c. 5%
    d. 10%

2. What is the largest gyre? (1)
    a. North Atlantic Subtropical Gyre
    b. North Pacific Subtropical Gyre
    c. South Atlantic Subtropical Gyre
    d. South Pacific Subtropical Gyre

3. Which of the following latitudes have the most pronounced upwelling? (1)
    a. 0
    b. 30 N
    c. 60 N
    d. 90 N

4. Geostrophic currents are balanced by pressure gradient force and what other force? (1)
    a. Gravity
    b. Coriolis Force
    c. Frictional force
    d. Wind stress

5. What are the major differences between western and eastern currents and what causes this difference? (3)

6. How does El Nino affect upwelling along the eastern Pacific, equatorial currents, and the thermocline in the Pacific? (3)

7. What is Langmuir Circulation and how can it be observed at the surface? (3)

Section 6: Tides (10)

1. To what power are tide-generating forces proportional to distance? (1)
    a. 0
    b. -1
    c. -2
    d. -3

2. How long is a tidal day? (1)
    a. 25 hours
    b. 24 hours 50 mins
    c. 24 hours 25 mins
    d. 23 hours 56 mins

3. How much of the Moon's tide-generating force does the Sun exert on Earth? (1)
    a. 25%
    b. 50%
    c. 100%
    d. 150%

4. Between about what latitudes do tidal bulges alternate? (1)
    a. 5 N and 5 S
    b. 10 N and 10 S
    c. 20 N and 20 S
    d. 30 N and 30 S

5. What is an amphidromic point and in what direction do tides rotate around these points? (3)

6. What causes tidal inequality and how does it affect the tidal pattern at a location? (3)

Section 7: Coastal Processes (9)

1. Which of the following is found the closest to the lagoon on a barrier island? (1)
    a. Barrier flat
    b. Dune
    c. Salt marsh
    d. Beach

2. Which of the following is an erosional feature? (1)
    a. Sea caves
    b. Tombolo
    c. Spit
    d. Barrier island

3. What is the pattern of movement of sediment by longshore current? (1)
    a. Straight
    b. Zigzag
    c. Curved
    d. Circular

4. Which of the following stabilization techniques do not cause increased erosion elsewhere? (1)
    a. Beach replenishment
    b. Groins
    c. Breakwaters
    d. Jetties

5. How do the formation of glaciers affect eustatic and isostatic adjustment and how does this affect the overall sea level? (3)

6. Where do waves normally refract toward and how does this affect coastlines over time? (2)

Section 8: Marine Sediments (10)

1. What is the most common type of marine sediment? (1)
    a. Lithogenous
    b. Biogenous
    c. Hydrogenous
    d. Cosmogenous

2. If primary productivity along the equator increases, which type of sediment will show the greatest increase in concentration? (1)
    a. Tektites
    b. Evaporites
    c. Calcareous Ooze
    d. Siliceous Ooze

3. Which of the following are not found in pelagic deposits? (1)
    a. Calcareous Ooze
    b. Siliceous Ooze
    c. Stromatoloites
    d. Manganese nodules

4. Which of the following types of deposition are most common for red clays? (1)
    a. Turbidity currents
    b. Suspension settling
    c. Hydrothermal deposition
    d. Ice-rafted debris

5. How does temperature affect the formation of calcareous ooze? And how does climate both impact the formation of calcareous ooze? (3)

6. How are manganese nodules formed on the seafloor? What marine structures are involved and commonly in what location? (3)

Section 9: Climate Change (9)

1. Which of the following is not a major effect of climate change? (1)
    a. Increased coastal erosion
    b. Decreased Earth albedo
    c. Increased ocean salinity
    d. Decreased carbonate deposition

2. Which of the following is most responsible for sea level rise from global warming? (1)
    a. Thermal expansion of seawater
    b. Increased volcanic activity
    c. Decreased precipitation
    d. Increased river discharge

3. Which of the following is not a direct effect of climate change on ocean chemistry? (1)
    e. Ocean acidification
    f. Increased hypoxia in deep waters
    g. Decreased salinity due to melting ice
    h. Increased primary productivity

4. How do melting polar ice caps affect thermohaline circulation? (3)

5. How does global warming affect coral ecosystems? What organism is primarily affected and through what process? (3)